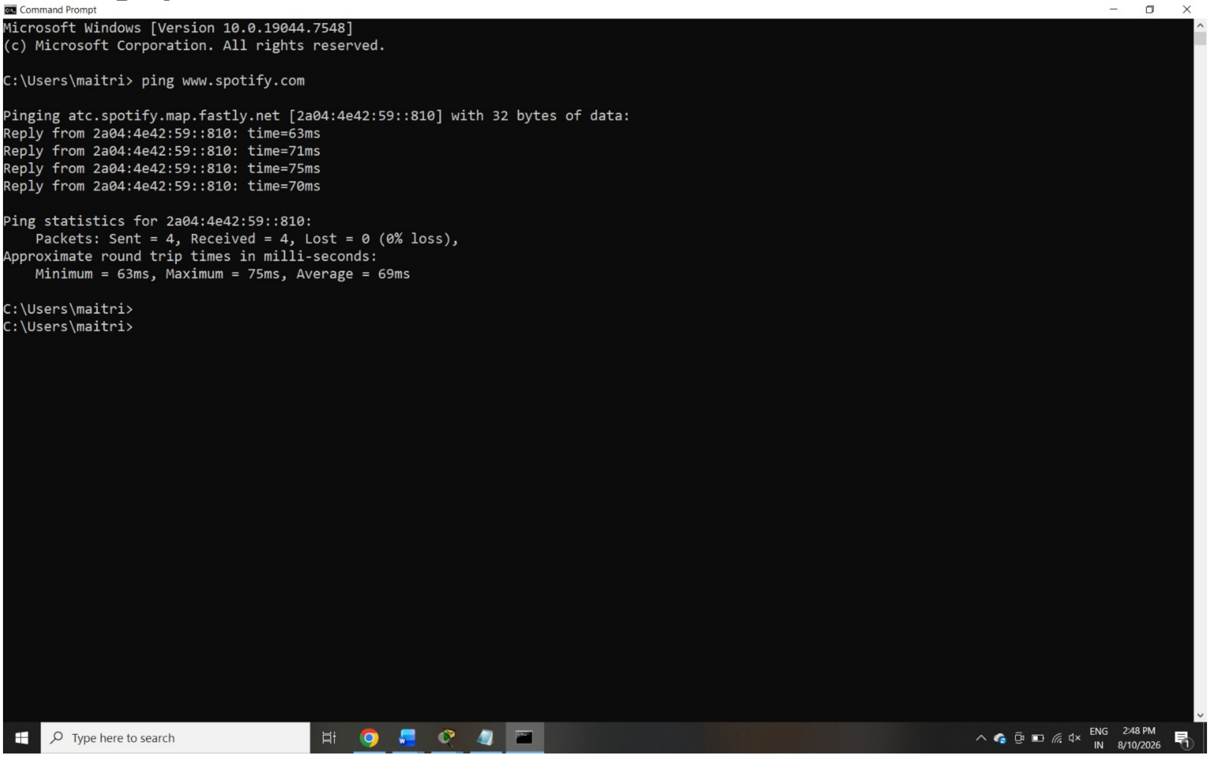
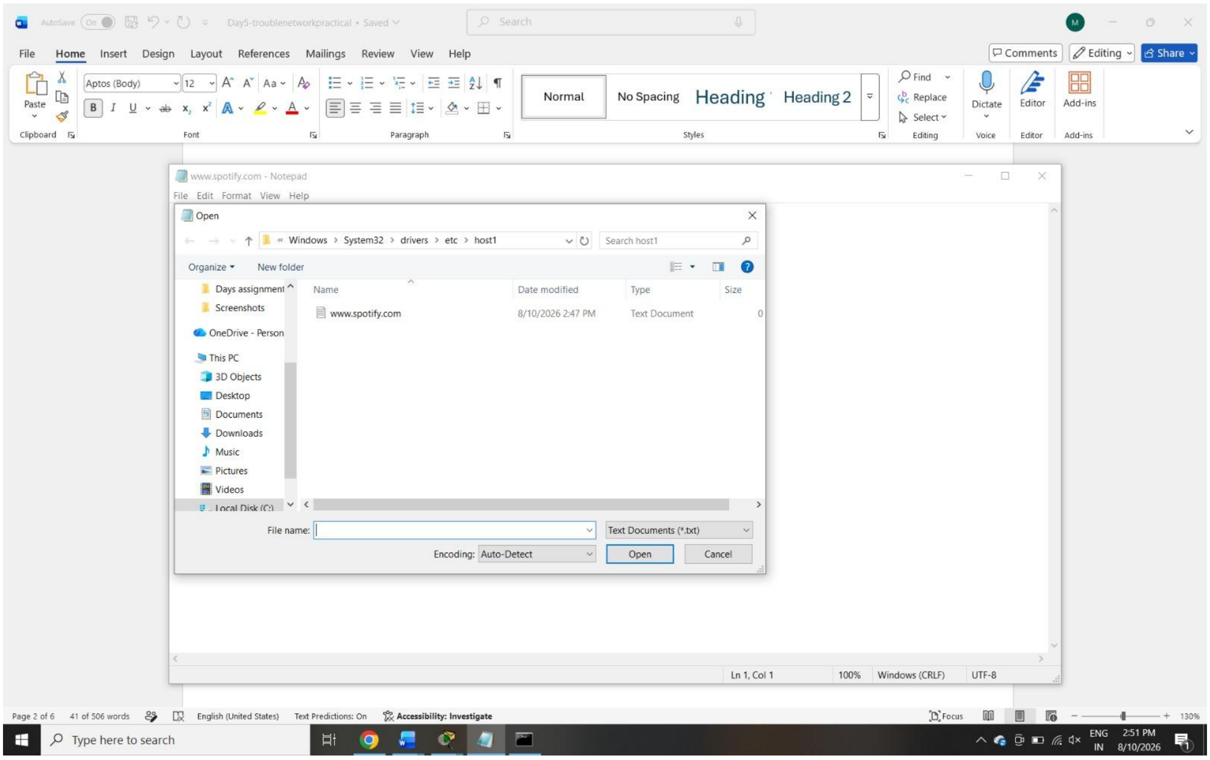
1.Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.





2.Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

DHCP (Dynamic Host Configuration Protocol) automatically provides network settings such as IP address, subnet mask, default gateway, and DNS server to client devices.

In this scenario, the DHCP server is intentionally configured with an **incorrect default gateway**. When a client requests an IP address, the DHCP server provides the IP address along with the wrong gateway.

For example, the client may receive:

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.254

But the actual router gateway is 192.168.1.1.

The client can communicate within the local network, but it may not be able to communicate with other networks or access the Internet because the default gateway is incorrect.

Troubleshooting:

1. Check the client's IP configuration.
2. Verify the default gateway using ipconfig.
3. Compare the gateway with the router's IP address.
4. Check the DHCP server configuration.
5. Correct the gateway address.
6. Renew the client's DHCP configuration.

3.Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

```
C:\Users\helLO>nslookup google.com
Server:  reliance.reliance
Address:  2405:201:2033:6025::c0a8:1d01

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4009:80e::200e
          192.178.211.100
          192.178.211.101
          192.178.211.102
          192.178.211.113
          192.178.211.138
          192.178.211.139


C:\Users\helLO>ping google.com

Pinging google.com [2404:6800:4009:80e::200e] with 32 bytes of data:
Reply from 2404:6800:4009:80e::200e: time=29ms
Reply from 2404:6800:4009:80e::200e: time=28ms
Reply from 2404:6800:4009:80e::200e: time=16ms
Reply from 2404:6800:4009:80e::200e: time=18ms

Ping statistics for 2404:6800:4009:80e::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 29ms, Average = 22ms
```

```
C:\Users\helLO>ping google.com

Pinging google.com [2404:6800:4009:80e::200e] with 32 bytes of data:
Reply from 2404:6800:4009:80e::200e: time=29ms
Reply from 2404:6800:4009:80e::200e: time=28ms
Reply from 2404:6800:4009:80e::200e: time=16ms
Reply from 2404:6800:4009:80e::200e: time=18ms

Ping statistics for 2404:6800:4009:80e::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 29ms, Average = 22ms

C:\Users\helLO>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=26ms TTL=112
Reply from 8.8.8.8: bytes=32 time=22ms TTL=112
Reply from 8.8.8.8: bytes=32 time=25ms TTL=112
Reply from 8.8.8.8: bytes=32 time=22ms TTL=112

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 22ms, Maximum = 26ms, Average = 23ms
```

4.A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

APIPA stands for **Automatic Private IP Addressing**. When a device is configured to obtain an IP address automatically but cannot communicate with the DHCP server, Windows may assign an IP address in the range:

169.254.x.x

For example:

169.254.10.25

An APIPA address indicates that the device did not receive a valid IP address from the DHCP server.

Possible Causes:

1. DHCP server is switched off.
2. DHCP service is not running.
3. DHCP scope is exhausted.
4. Network cable is disconnected.
5. Switch port has a problem.
6. Wrong VLAN configuration.
7. DHCP server is unreachable.
8. DHCP relay is incorrectly configured.

Troubleshooting:

First, check the physical network connection and make sure the device is properly connected to the switch.

Next, check whether the DHCP server is running and whether DHCP is enabled.

Then check the DHCP scope to make sure there are enough available IP addresses.

After that, verify the VLAN and network configuration. If the DHCP server is located on another network, check the DHCP relay configuration.

Finally, renew the IP address using:

ipconfig /release
ipconfig /renew

The device should receive a valid IP address such as 192.168.1.x.

5.Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

DNS Issue	Symptoms	Possible Cause	Solution
DNS Server Unavailable	Websites do not open by name	DNS server is down	Check or configure a working DNS server
Incorrect DNS Configuration	Website works by IP but not by name	Wrong DNS server address	Check DNS settings
DNS Cache Problem	Incorrect or old IP is returned	Outdated DNS cache	Use ipconfig /flushdns

DHCP Issue	Symptoms	Possible Cause	Solution
DHCP Server Unavailable	Device gets 169.254.x.x	DHCP server is off/unavailable	Check and enable DHCP server
DHCP Scope Exhausted	New devices cannot get IP	All available IPs are used	Increase DHCP scope or free IP addresses
Incorrect Default Gateway	Internet/other networks cannot be accessed	Wrong gateway supplied by DHCP	Correct the DHCP gateway